# João André **Roque Costa**

Software, Data & AI Engineer
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Computer Science and Engineering graduate focused on **Artificial Intelligence**, **Data Science**, and **Python** development. Built data and automation prototypes with **Azure AI Foundry**, **Azure Data Factory**, Databricks, RAG, and OCR, and developed a parallelizable thesis method for nonlinear dimensionality reduction on large-scale datasets. Brings practical experience integrating APIs and data platforms, building monitoring-oriented self-hosted tools, and communicating technical work across teams and audiences.

## Professional Experience

Tech Analyst — *Coimbra, Portugal*
**Deloitte** — *Apr. 2026 - Oct. 2026*

- Delivered a resource-group migration, including **Databricks** and **Azure Data Factory** assets, to a new Azure subscription within the planned timeframe.
- Contributed to a cross-functional **LLM** prototype for data migration using **Azure AI Foundry**, translating a data-centric business need into an AI-enabled workflow.
- Built a working **Python** and **Playwright** proof of concept for automating HR report generation.
- Contributed technical context to a client-facing proposal for documentation automation using **RAG** and **OCR**.
- Worked with **Azure Blob Storage**, Azure Managed Instances, MongoDB, MySQL, Git, and Databricks while ramping up on an existing cloud project structure.

Teaching Assistant — *Lisbon, Portugal*
**Instituto Superior Técnico** — *Oct. 2024 - Jul. 2025*

- Lectured and evaluated practical classes, tracked student progress, and assisted projects involving **Python**, pandas, matplotlib, data treatment, visualization, and analysis.
- Adapted explanations and feedback to different audiences while mentoring students through practical data-science work.

## Education

M.Sc. Computer Science and Engineering — *Lisbon, Portugal*
**Instituto Superior Técnico** — *Sep 2023 - Nov 2025*

- M.Sc. in Computer Science and Engineering, with a self-directed focus on **Artificial Intelligence** and **Data Science**; final grade 15/20.
- Relevant Coursework: **Deep Learning**, Machine Learning, Data Science, Data Analysis and Integration, Cloud Computing and Virtualization, Optimization Algorithms, Planning, Learning and Intelligent Decision Making, Autonomous Agents and Multi-Agent Systems
- Thesis: Maximum Variance Unfolding on Disjoint Manifolds (MVU-DM)
  - Developed a decomposition-based extension to MVU for nonlinear dimensionality reduction on disconnected datasets.
  - Designed a parallelizable approach that improved scalability for large-scale datasets while preserving local geometric structure in comparable embeddings.
  - Used **Python**, NumPy, and machine-learning evaluation to develop and assess the method.

B.Sc. Computer Science and Engineering — *Lisbon, Portugal*
**Instituto Superior Técnico** — *Sep 2020 - Jul 2023*

- Relevant Areas: **Algorithms**, data structures, distributed systems, operating systems, database theory, and systems programming
  - Applied algorithmic analysis, data structures, complexity, and design in the Analysis and Synthesis of Algorithms coursework.
  - Studied scalability, fault tolerance, security, remote calls, and distributed naming in Distributed Systems coursework.
  - Worked with C, C++, Java, Python, PostgreSQL, Linux, Bash scripting, and RESTful APIs across academic projects.

## Extracurricular

- Built **Autolab**, a self-hosted modular system on a Linux NUC with per-machine resource management, product price tracking, and a virtual dashboard for on-the-fly monitoring.
- Developed a Discord-based matchmaking application integrating the **Discord API**, PostgreSQL, and Redis for ELO ranking, match history, caching, and queue management.
- Built price-tracking and marketplace automation tools using **Python**, Selenium, BeautifulSoup, web interfaces, and multi-platform API integration.

## Skills & Competencies

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| **AI / ML** | **LLMs**, RAG, OCR, Machine Learning, Deep Learning, TensorFlow, Keras, Scikit-learn, Multi-Agent Systems, nonlinear dimensionality reduction |
| **Python & APIs** | **Python**, NumPy, Pandas, Matplotlib, Playwright, Selenium, BeautifulSoup, RESTful APIs, Discord API |
| **Cloud & Data** | **Azure AI Foundry**, Azure Data Factory, Azure Blob Storage, Azure Managed Instances, Databricks |
| **Databases & Infra** | PostgreSQL, MySQL, MongoDB, Redis, Docker, Linux, Bash scripting, Git |
| **Communication** | technical writing, lecturing, mentoring, evaluation, stakeholder communication |